

LCoffee

April 19, 2026

Outline

① Updates

② Mechanism

Two Outcome Variables

The four update regressions use two families of outcome variables Y_{iet} :

Outcome type	Definition
Purchase frequency	$n_purchases$ = purchases in the event-time window divided by window length (daily rate)
Variety seeking	share of purchases that are “new” (never purchased before) within the window

- Purchase-frequency slides ask whether closures change *how much* members buy afterward
- Variety-seeking slides ask whether closures change *what kinds of products* members explore afterward

Results

- Closures do not provide strong evidence of lasting habit disruption in purchase frequency.
 - The large negative post x disp term indicates the existence of a purchase circle.
 - The positive purchase DDD displacement effect indicates a stronger returning intention driven by forced-no-purchase, instead of forgetting.
- Closures appear to affect product choice more than total quantity purchased.
 - In the variety DDD, higher displacement propensity reduces the increase in exploration.

Update 1: Matched Purchase DiD

Matched panel of members observed in every pre/post period, with Y_{iet} = daily purchase rate and $t \in \{-4, -3, -2, -1, 1, 2, 3, 4\}$:

$$Y_{iet} = \delta \cdot \mathbf{1}(\text{post}_t) \times \mathbf{1}(\text{Treated}_{ie}) + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}$$

Parameter of interest

$$\hat{\delta} = 0.00514 \text{ (0.00441)}$$

Result

- $p = 0.244$: insignificant
- Treated / control members: 952 / 4,248
- 22 closures, 5,200 members, 41,600 obs.
- Pre-trend joint test: $p = 0.458$

Update 2: Purchase DDD

Unmatched-panel DDD for purchase frequency, where D_{ie} is the displacement indicator:

$$Y_{iet} = \delta^B(\text{post}_t \times \text{Treated}_{ie}) + \beta(\text{post}_t \times D_{ie}) \\ + \delta^D(\text{post}_t \times \text{Treated}_{ie} \times D_{ie}) + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}$$

Estimated parameters

Parameter	Estimate (SE)
$\hat{\delta}^B$	0.00081 (0.00064)
$\hat{\beta}$	-0.03353*** (0.00107)
$\hat{\delta}^D$	0.00517** (0.00241)

Result

- Treated / control members: 8,092 / 36,766
- Baseline effect is near zero
- Displacement effect is small and positive
- Pre-trend didn't pass $p = 0.004$

Update 3: Matched Variety-Seeking DiD

Matched-panel DiD for variety seeking:

$$Y_{iet} = \delta \cdot \mathbf{1}(\text{post}_t) \times \mathbf{1}(\text{Treated}_{ie}) + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}$$

Estimated treatment effect

Outcome mode	$\hat{\delta}$
Distinct products	0.04066*** (0.01419)
Purchase instances	0.02991** (0.01435)

Result

- Treated / control members: 308 / 1,415
- Positive post-closure effect in both outcome constructions
- Pre-trends supported: $p = 0.525$ and 0.686

Update 4: Variety-Seeking DDD

Unmatched-panel DDD for distinct-mode variety seeking:

$$Y_{iet} = \delta^B(\text{post}_t \times \text{Treated}_{ie}) + \beta(\text{post}_t \times D_{ie}) \\ + \delta^D(\text{post}_t \times \text{Treated}_{ie} \times D_{ie}) + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}$$

Estimated parameters

Parameter	Estimate (SE)
$\hat{\delta}^B$	0.031*** (0.011)
$\hat{\beta}$	0.036*** (0.006)
$\hat{\delta}^D$	-0.039*** (0.014)

Result

- Treated / control members: 8,092 / 36,766
- Baseline effect is positive for non-displaced treated members
- Displacement effect is negative

Mechanism 1: Goal Carryover

Idea

- Some customers intended to buy during the closure, but were temporarily blocked.
- After reopening, these unsatisfied purchase goals generate a short-run rebound in purchase prob.

Supportive evidence 1: Positive and significant immediately after reopening

Week	δ_ℓ	p
-4	0.00043	0.720
-3	-0.00028	0.809
-2	-0.00031	0.759
1	0.00254**	0.014
2	0.00016	0.884
3	0.00623***	< 0.001
4	0.00175	0.130

Supportive evidence 2

- 7.3% of selected treatment members purchased at another store during the closure.

Mechanism 2: Missed Exposure to New Products

Idea

- Control customers keep purchasing during the closure, so they continue to see and try newly introduced products.
- Treated high-intention customers are interrupted and miss this exposure window.
- After reopening, treated customers may be less familiar with the updated assortment and return to safer, previously chosen products.
- This can lower post-closure novelty-seeking for displaced treated customers relative to similarly high-intention controls.

Interpretation

- The negative variety-seeking DDD effect may partly reflect missed exposure to assortment changes during the closure period.

Mechanism 2: Potential Tests

Descriptive evidence

- Across 22 closures, the matched control stores introduced on average 7.9 new products per store during the closure.
- The median average is 8.6, with a range from 1.0 to 15.4.

Potential tests

- Test whether the negative variety-seeking displacement effect is stronger when matched control stores experienced more new-product introductions during the closure.
- Test whether treated customers who purchased at other stores during the closure show a weaker decline in novelty-seeking.
- Use product introduction timing to test whether treated customers are less likely to adopt products first introduced during the closure period.